

Python for Data Science - 2305CS303

Lab - 6

Roll No. : 128

Name : Riyaz

01) WAP to find sum of tuple elements.

Sample Input:

T = (10, 20, 30, 40)

Sample Output:

Sum of tuple elements = 100

```
In [4]: T=(10,20,30,40)
        sum(T)
```

```
Out[4]: 100
```

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

T = (7, 2, 9, 4, 1, 5)
K = 2

Sample Output:

Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)

```
In [19]: t=(7, 2, 9, 4, 1, 5)
l=list(sorted(t))
print(l)
mi=list(l[:min(l)+1])
ma=list(l[-1:-3:-1])
print("Minimum 2 elements: ",tuple(mi))
print("Maximum 2 elements: ",tuple(ma))
```

```
[1, 2, 4, 5, 7, 9]
Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)
```

03) WAP to find tuples which have all elements divisible by K from a list of tuples.

Sample Input:

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

```
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

```
In [20]: tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
result=[]
for i in tuple_list:
    a=True
    for j in i:
        if j % 2!=0:
            a=False
            break
    else:
        result.append(i)
result
```

```
Out[20]: [(2, 4, 6), (4, 8, 12)]
```

04) WAP to create a list of tuples from given list having number and its cube in each tuple.

Sample Input:

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]
```

```
In [22]: numbers = [1, 2, 3, 4, 5]
result=[]
for i in numbers:
    j=(i,i**3)
    result.append(j)
result
```

```
Out[22]: [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]
```

05) WAP to find tuples with all positive elements from the given list of tuples.

Sample Input:

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
```

Sample Output:

```
Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]
```

```
In [24]: tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
result=[]
for i in tuple_list:
    for j in i:
        if j<=0:
            break
    else:
        result.append(i)
result
```

```
Out[24]: [(1, 2, 3), (4, 5, 6)]
```

06) WAP to remove tuples of length K.

Sample Input:

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

Sample Output:

```
List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8, 9)]
```

```
In [26]: tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
k = 2
result=[]
for i in tuple_list:
    if len(i)!=k:
```

```
        result.append(i)
result
```

Out[26]: [(3, 4, 5), (6,), (7, 8, 9)]

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

(5, 7, 9)

```
In [27]: T = (3, 5, 3, 7, 3, 9)
x = 3
t=()
for i in T:
    if i!=x:
        t+=(i,)
t
```

Out[27]: (5, 7, 9)

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

(3, 5, 7, 9)

```
In [29]: T = (3, 5, 3, 7, 3, 9)
s=set(T)
print(tuple(s))
```

(9, 3, 5, 7)

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

```
t = (2, 3, 4, 5)
```

Sample Output:

Resultant tuple after multiplying adjacent elements: (6, 12, 20)

```
In [31]: t = (2, 3, 4, 5)
T=()
for i in range(len(t)-1):
    T+=(t[i]*t[i+1],)
T
```

Out[31]: (6, 12, 20)

10) WAP to test if the given tuple is distinct or not.

Sample Input:

```
tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
```

Sample Output:

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False
```

```
In [32]: tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
if len(tuple(set(tuple1)))==len(tuple1):
    print("tuple1 = (1, 2, 3, 4) is distinct? -> True")
else:
    print("tuple = (1, 2, 3, 4) is distinct? -> False")
if len(tuple(set(tuple2)))==len(tuple2):
    print("tuple2 = (1, 2, 3, 4) is distinct? -> True")
else:
    print("tuple2 = (1, 2, 3, 4) is distinct? -> False")
```

```
tuple1 = (1, 2, 3, 4) is distinct? -> True
tuple2 = (1, 2, 3, 4) is distinct? -> False
```

11) WAP to rotate the elements of a tuple to the right by **k** positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

```
In [36]: T = (10, 20, 30, 40, 50)
k = 2
l=T[k+1:]+T[:k+1]
l
```

Out[36]: (40, 50, 10, 20, 30)

12) WAP to merge two tuples of equal length alternately.

Sample Input:

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

```
In [39]: T1 = (1, 3, 5)
T2 = (2, 4, 6)
if len(T1)==len(T2):
    T3=(T1+T2)
    print(tuple(sorted(T3)))
else:
    print("Tuple length are not equal")
```

```
(1, 2, 3, 4, 5, 6)
```

13) WAP to create a tuple of squares of elements that are even and greater than 10.

Sample Input:

```
T = (4, 12, 7, 18, 10)
```

Sample Output:

```
(144, 324)
```

```
In [42]: T = (4, 12, 7, 18, 10)
t=[]
for i in T:
    if i %2==0:
        if i>10:
            t.append(i**2)
tuple(t)
```

```
Out[42]: (144, 324)
```

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

```
T = (4, 7, 9, 11, 15)
```

Sample Output:

```
((1, 7), (3, 11))
```

```
In [63]: def prime(l):
    for j in range(2,l):
        if l%j==0:
            return False
        else:
            return True

T = (4, 7, 9, 11, 15)
t=()
for i in range(len(T)):
    b=prime(T[i])
    if b:
        k=tuple([i,T[i]])
        t+=(k,)
t
```

Out[63]: ((1, 7), (3, 11))

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

```
T = ('a', 'b', 'c', 'd', 'e')
```

Sample Input:

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

```
In [66]: T = ('a', 'b', 'c', 'd', 'e')
even=()
odd=()
for i in range(len(T)):
    if i%2==0:
        even+=(T[i],)
    else:
        odd+=(T[i],)
print(even)
print(odd)
```

```
('a', 'c', 'e')
('b', 'd')
```

16) WAP to compute column-wise sum from a tuple of lists.

Sample Input:

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
```

Sample Output:

```
12 15 18
```

```
In [78]: T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
result=()
for i in range(len(T)):
    total=0
    for j in range(len(T[i])):
        total+=T[j][i]
    result+=(total,)
print(result)
```

(12, 15, 18)